Atiyah Macdonald exercises

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1 Rings and Ideals

Exercise 1.12.

1. $\mathfrak{a} \subseteq (\mathfrak{a} : \mathfrak{b})$
2. $(\mathfrak{a} : \mathfrak{b})\mathfrak{b} \subseteq \mathfrak{a}$
3. $((\mathfrak{a} : \mathfrak{b}) : \mathfrak{c}) = (\mathfrak{a} : \mathfrak{b}\mathfrak{c}) = ((\mathfrak{a} : \mathfrak{c}) : \mathfrak{b})$
4. $(\bigcap_i \mathfrak{a}_i : \mathfrak{b}) = \bigcap_i (\mathfrak{a}_i : \mathfrak{b})$
5. $(\mathfrak{a} : \sum_i \mathfrak{b}_i) = \bigcap_i (\mathfrak{a} : \mathfrak{b}_i)$.

Solution:

1. For $a \in \mathfrak{a}$, $a\mathfrak{b} \subseteq \mathfrak{a}$ since $\mathfrak{a}$ is an ideal.
2. Since $xb \in (\mathfrak{a} : \mathfrak{b})\mathfrak{b}$ iff $x\mathfrak{b} \subseteq \mathfrak{a}$ and $b \in \mathfrak{b}$, $xb \in \mathfrak{a}$.
3. Since $\mathfrak{b}\mathfrak{c} = \mathfrak{c}\mathfrak{b}$ we only need to show the first equality. Observe that $x \in ((\mathfrak{a} : \mathfrak{b}) : \mathfrak{c})$ iff $x\mathfrak{c} \subseteq (\mathfrak{a} : \mathfrak{b})$, that is for all $xc \in x\mathfrak{c}$ with $c \in \mathfrak{c}$, $xc\mathfrak{b} \subseteq \mathfrak{a}$. This is true iff $x\mathfrak{c}\mathfrak{b} \subseteq \mathfrak{a}$.
4. $x\mathfrak{b} \subseteq \bigcap_i \mathfrak{a}_i$ iff $x\mathfrak{b} \subseteq \mathfrak{a}_i$ for all i.
5. Note $x \sum_i \mathfrak{b}_i = \sum_i x\mathfrak{b}_i$. Then $\sum_i x\mathfrak{b}_i \subseteq \mathfrak{a}$ iff $x\mathfrak{b}_i \subseteq \mathfrak{a}$ for all i.

Exercise 1.13.

1. $\sqrt{\mathfrak{a}} \supseteq \mathfrak{a}$
2. $\sqrt{\sqrt{\mathfrak{a}}} = \sqrt{\mathfrak{a}}$
3. $\sqrt{\mathfrak{a}\mathfrak{b}} = \sqrt{\mathfrak{a} \cap \mathfrak{b}} = \sqrt{\mathfrak{a}} \cap \sqrt{\mathfrak{b}}$
4. $\sqrt{\mathfrak{a}} = (1) \iff \mathfrak{a} = (1)$
5. $\sqrt{\mathfrak{a} + \mathfrak{b}} = \sqrt{\sqrt{\mathfrak{a}} + \sqrt{\mathfrak{b}}}$
6. if $\mathfrak{p}$ is prime, $\sqrt{\mathfrak{p}^n} = \mathfrak{p}$ for all $n > 0$.

Solution:

1. If $x \in \mathfrak{a}$, then $x^1 \in \mathfrak{a}$ so $x \in \sqrt{\mathfrak{a}}$.

2. $x \in \sqrt{\sqrt{\mathfrak{a}}}$ iff $(x^n)^m \in \mathfrak{a}$ iff $x^{nm} \in \mathfrak{a}$ iff $x \in \sqrt{\mathfrak{a}}$.

3. If $x^n \in \mathfrak{ab}$ then $x^n \in \mathfrak{a} \cap \mathfrak{b}$. If $x^n \in \mathfrak{a} \cap \mathfrak{b}$ then $x^n \in \mathfrak{a}$ and $x^n \in \mathfrak{b}$. If $x^n \in \mathfrak{a}$ and $x^m \in \mathfrak{b}$ then $x^{mn} \in \mathfrak{ab}$.

4. If $1^n \in \mathfrak{a}$ then $1 \in \mathfrak{a}$. The converse comes from 1.

5. By 1., $\sqrt{\mathfrak{a}+\mathfrak{b}} \subseteq \sqrt{\sqrt{\mathfrak{a}}+\sqrt{\mathfrak{b}}}$. If $x^n = y+z$ where $y^k \in \mathfrak{a}$ and $z^m \in \mathfrak{b}$ then $x^{n(k+m)} = (y+z)^{k+m}$ has every term in its binomial expansion with either its y term at least k or its z term at least m, and so $x^{n(k+m)} \in \mathfrak{a}+\mathfrak{b}$.

6. If $x^m \in \mathfrak{p}^n$ then $x^m \in \mathfrak{p}$ so $x \in \mathfrak{p}$. Conversely if $x \in \mathfrak{p}$ then $x^n \in \mathfrak{p}^n$ so $x \in \sqrt{\mathfrak{p}^n}$.

Exercise 1.18. If $\mathfrak{a}_1, \mathfrak{a}_2$ are ideals of A and if $\mathfrak{b}_1, \mathfrak{b}_2$ are ideals of B, then

$$\begin{array}{cc}
(\mathfrak{a}_1+\mathfrak{a}_2)^e = \mathfrak{a}_1^e + \mathfrak{a}_2^e, & (\mathfrak{b}_1+\mathfrak{b}_2)^c \supseteq \mathfrak{b}_1^c + \mathfrak{b}_2^c, \\
(\mathfrak{a}_1 \cap \mathfrak{a}_2)^e \subseteq \mathfrak{a}_1^e \cap \mathfrak{a}_2^e, & (\mathfrak{b}_1 \cap \mathfrak{b}_2)^c = \mathfrak{b}_1^c \cap \mathfrak{b}_2^c, \\
(\mathfrak{a}_1\mathfrak{a}_2)^e = \mathfrak{a}_1^e\mathfrak{a}_2^e, & (\mathfrak{b}_1\mathfrak{b}_2)^c \supseteq \mathfrak{b}_1^c\mathfrak{b}_2^c, \\
(\mathfrak{a}_1 : \mathfrak{a}_2)^e \subseteq (\mathfrak{a}_1^e : \mathfrak{a}_2^e), & (\mathfrak{b}_1 : \mathfrak{b}_2)^c \subseteq (\mathfrak{b}_1^c : \mathfrak{b}_2^c), \\
\sqrt{\mathfrak{a}}^e \subseteq \sqrt{\mathfrak{a}^e} & (\sqrt{\mathfrak{b}})^c = \sqrt{\mathfrak{b}^c}.
\end{array}$$

Solution: Note that extension is monotone. Hence $\mathfrak{a}_1^e \subseteq (\mathfrak{a}_1+\mathfrak{a}_2)^e$ and $\mathfrak{a}_2^e \subseteq (\mathfrak{a}_1+\mathfrak{a}_2)^e$. Hence $\mathfrak{a}_1^e + \mathfrak{a}_2^e \subseteq (\mathfrak{a}_1+\mathfrak{a}_2)^e$. Conversely, let $\sum_i f(a_i+b_i) \in (\mathfrak{a}_1+\mathfrak{a}_2)^e$. Then $\sum_i f(a_i+b_i) = (\sum_i f(a_i)) + (\sum_i f(b_i)) \in \mathfrak{a}_1^e + \mathfrak{a}_2^e$. Next, notice that $(\mathfrak{a}_1 \cap \mathfrak{a}_2)^e \subseteq \mathfrak{a}_1^e$ and $(\mathfrak{a}_1 \cap \mathfrak{a}_2)^e \subseteq \mathfrak{a}_2^e$. Hence $(\mathfrak{a}_1 \cap \mathfrak{a}2)^e \subseteq \mathfrak{a}_1^e \cap \mathfrak{a}_2^e$. Next, let $\sum_i f(a_ib_i) \in (\mathfrak{a}_1\mathfrak{a}_2)^e$. Then $\sum_i f(a_ib_i) = \sum_i f(a_i)f(b_i) \in \mathfrak{a}_1^e\mathfrak{a}_2^e$. Conversely let $\sum_i(\sum_j f(a_{ij}))(\sum_k f(b_{ik})) \in \mathfrak{a}_1^e\mathfrak{a}_2^e$. Then $\sum_i(\sum_j f(a_{ij}))(\sum_k f(b_{ik})) = \sum_i \sum_j \sum_k f(a_{ij})f(b_{ik}) = \sum_i \sum_j \sum_k f(a_{ij}b_{ik}) \in (\mathfrak{a}_1\mathfrak{a}_2)^e$. Next, let $\sum_i f(a_i) \in (\mathfrak{a}_1 : \mathfrak{a}_2)^e$ such that $a_i\mathfrak{a}_2 \subseteq \mathfrak{a}_1$. Then $(f(a_i))\mathfrak{a}_2^e = (a_i)^e\mathfrak{a}_2^e = (a_i\mathfrak{a}_2)^e \subseteq a_1^e$ so $f(a_i) \in (\mathfrak{a}_1^e : \mathfrak{a}_2^e)$. Next let $\sum_i f(a_i) \in \sqrt{\mathfrak{b}}^e$ Then for some n_i, $f(a_i)^{n_i} \in \mathfrak{a}^e$ so $\sum_i f(a_i) \in \sqrt{\mathfrak{a}^e}$.

Let $a+b \in \mathfrak{b}_1^c + \mathfrak{b}_2^c$. Then $f(a) \in \mathfrak{b}_1, f(b) \in \mathfrak{b}_2$ so $f(a+b) \in \mathfrak{b}_1 + \mathfrak{b}_2$ hence $a+b \in (\mathfrak{b}_1+\mathfrak{b}_2)^c$. Next let $b \in (\mathfrak{b}_1 \cap \mathfrak{b}_2)^c$. Then $f(b) \in \mathfrak{b}_1 \cap \mathfrak{b}_2$. Hence $b \in \mathfrak{b}_1^c \cap \mathfrak{b}_2^c$. Conversely if $b \in \mathfrak{b}_1^c \cap \mathfrak{b}_2^c$ then $f(b) \in \mathfrak{b}_1$ and $f(b) \in \mathfrak{b}_2$ so $b \in (\mathfrak{b}_1 \cap \mathfrak{b}_2)^c$. Next let $\sum_i a_ib_i \in \mathfrak{b}_1^c\mathfrak{b}_2^c$. Then $f(a_i) \in \mathfrak{b}_1$ and $f(b_i) \in \mathfrak{b}_2$ so $f(\sum_i a_ib_i) \in \mathfrak{b}_1\mathfrak{b}_2$. Hence $\sum_i a_ib_i \in (\mathfrak{b}_1\mathfrak{b}_2)^c$. Next let $b \in (\mathfrak{b}_1 : \mathfrak{b}_2)^c$. Then $f(b)\mathfrak{b}_2 \subseteq \mathfrak{b}_1$, that is for each $c \in \mathfrak{b}_2$, $f(b)c \in \mathfrak{b}_1$. If $c' \in \mathfrak{b}_2^c$ so that $f(c') = c$ then $f(bc') \in \mathfrak{b}_1$, so $bc' \in \mathfrak{b}_1^c$. Hence $b \in (\mathfrak{b}_1^c : \mathfrak{b}_2^c)$. Next, if $b \in (\sqrt{\mathfrak{b}})^c$ so that $f(b)^n \in \mathfrak{b}$, then $f(b^n) \in \mathfrak{b}$ so $b^n \in \mathfrak{b}^c$, so $b \in \sqrt{\mathfrak{b}^c}$. Conversely, if $b \in \sqrt{\mathfrak{b}^c}$ so that $b^n \in \mathfrak{b}^c$, then $f(b)^n \in \mathfrak{b}$ so $f(b) \in \sqrt{\mathfrak{b}}$ so $b \in (\sqrt{\mathfrak{b}})^c$.

Exercise (1). Let x be a nilpotent element of a ring A. Show that $1+x$ is a unit in A. Deduce that the sum of a nilpotent element and a unit is a unit.

Solution: Suppose $x^n = 0$. Notice that

$$(1+x)(\prod_{i=0}^{n}(-1)^i x^i) = 1 + x - x - x^2 + x^2 + x^3 \cdots \pm x^n = 1 \pm x^n = 1.$$

If u is a unit then $u^{-1}x$ is nilpotent, so $1 + u^{-1}x$ is a unit, hence $u + x$ is.

Exercise (2). Let A be a ring and let $A[x]$ be the ring of polynomials in an indeterminate x, with coefficients in A. Let $f = a_0 + a_1x + \cdots + a_nx^n$. Prove that

1. f is a unit $\iff$ a_0 is a unit in A and $a_1, \cdots, a_n$ are nilpotent.
2. f is nilpotent $\iff$ $a_0, a_1, \cdots, a_n$ are nilpotent.
3. f is a zero-divisor $\iff$ there exists $a \neq 0$ in A such that $af = 0$.
4. f is said to be primitive if $(a_0, a_1, \cdots, a_n) = (1)$. Prove that if $f, g \in A[x]$, then fg is primitive $\iff$ f and g are primitive.

Solution:

1. If a_0 is a unit and a_i are nilpotent for $i \geq 1$, then $a_i x^i$ are nilpotent and their sum is nilpotent, hence by (1) f is a unit.

   Conversely suppose f is a unit with inverse g (coefficients indexed with b_i). Then $a_0 b_0 = 1$. Next let $\mathfrak{p}$ be a prime ideal. The map $\varphi : A[x] \to (A/\mathfrak{p})[x]$ has $\varphi(f)\varphi(g) = 1$. Hence the degree of $\varphi(f)$ and $\varphi(g)$ are both 0 since $(A/\mathfrak{p})[x]$ is an integral domain. Thus $\phi(a_i) = 0$ for all $i \geq 1$, that is $a_i \in \mathfrak{p}$. Since this is for any prime ideal, the a_i are in all prime ideals, and hence are nilpotent.

2. Similarly to above, if the a_i are nilpotent, their product with indeterminates are, and their sum is.

   Let f be nilpotent with $f^n = 0$. Then $a_0^n = 0$ so a_0 is nilpotent. By exercise (1), $f + 1$ is a unit, and by part 1 the a_i are nilpotent for $i \geq 1$.

3. Let $g \neq 0$ annihilate f with minimum degree m. Then $b_m a_n = 0$. Hence $a_n g$ has degree less than m, and moreover $a_n g$ annihilates f. Hence $a_n g = 0$. Now suppose for induction that $a_{n-i} g = 0$ holds for all $i \leq r$. Then $a_{n-r-1} b_i = -\sum_{j>i} a_{n-j} b_j = 0$. Hence for all i, $a_i b_n = 0$. Thus $b_n f = 0$.

4. Suppose fg is primitive, and call its coefficients c_i. It is sufficient to show f is primitive. Notice that $(a + b) \subseteq (a) + (b)$. Hence $(c_i) \subseteq \sum_{j \leq i}(a_j b_{i-j}) \subseteq \sum_{j \leq i}(a_j)$ for all i. Thus $(1) = \sum_i (c_i) \subseteq \sum_i (a_j)$, and so f is primitive.

   Conversely suppose f and g are primitive. We will show that $\mathfrak{c} = (c_0, c_1, \cdots, c_k)$ is not contained in any prime ideal of A. Suppose it were contained $\mathfrak{p}$. Then under the map $\varphi : A[x] \to (A/\mathfrak{p})[x]$, $\varphi(c_i) = 0$ and so $\varphi(fg) = 0$. Since $(A/\mathfrak{p})[x]$ is a domain, either $\varphi(f) = 0$ or $\varphi(g) = 0$. However, since $(a_0, a_1, \cdots, a_n) \not\supseteq \mathfrak{p}$ (since $\mathfrak{p}$ is proper), some a_i is not sent to zero. Similarly, some b_i is not sent to zero. This is a contradiction, and so $\mathfrak{c}$ is not contained in any prime ideal. Since every proper ideal is contained in a maximal (and hence prime) ideal, $\mathfrak{c}$ is not proper.

Exercise (3). Generalise the results in Exercise 1 to a polynomial ring $A[x_1, \cdots, x_n]$ in several indeterminates.

Solution: The proofs of part 1, 2, and 4 do not depend on being a single variable, so their proofs can be adapted exactly. We will only show part 3. Let f be a zero-divisor with annihilator g. Let d be greater than the maximum exponent of indeterminates in f. Consider the map $\varphi : A[x_1, \cdots, x_n] \to A[y]$ sending x_i to y^{d^i}. Notice that φ restricted to the set of polynomials with maximum exponent indeterminates less than d is injective. To see this, write the exponent of the image of each monomial in base d, each monomial is mapped distinctly. Then $\varphi(fg) = 0$ so $\varphi(f)$

is a zero-divisor. Hence there is an $a \neq 0$ such that $\varphi(af) = 0$. By injectivity of the restriction, $af = 0$.

Exercise (4). In the ring $A[x]$, the Jacobson radical is equal to the nilradical.

Solution: Since the set of maximal ideals is a subset of the set of prime ideals, the nilradical is contained in the Jacobson radical. Conversely let f be in the Jacobson radical. Then $xf - 1$ is a unit, so by Exercise 1 all coefficients of f are nilpotent. Hence f is nilpotent.

Exercise (5). Let A be a ring and let $A[[x]]$ be the ring of formal power series $f = \sum_{n=0}^{\infty} a_n x^n$ with coefficients in A. Show that

1. f is a unit in $A[[x]] \iff a_0$ is a unit in A.
2. If f is nilpotent, then a_n is nilpotent for all $n \geq 0$. Is the converse true?
3. f belongs to the Jacobson radical of $A[[x]] \iff a_0$ belongs to the Jacobson radical of A.
4. The contraction of a maximal ideal $\mathfrak{m}$ of $A[[x]]$ is a maximal ideal of A, and $\mathfrak{m}$ is generated by $\mathfrak{m}^c$ and x.
5. Every prime ideal of A is the contraction of a prime ideal of $A[[x]]$.

Solution:

1. If f is a unit with inverse $g = \sum_{n=0}^{\infty} b_n x^n$ then $1 = a_0 b_0$ so a_0 is a unit.

   Conversely, suppose a_0 is a unit. We define the inverse g of f by induction. Let b_0 be the inverse of a_0. Next, assume we have the first n terms of g constructed so that the first n terms in their product is zero. Then $c_n + 1 = \sum_{i=0}^{n+1} a_i b_{n+1-i}$. We want $c_n + 1 = 0$ so we get that $b_{n+1} a_0 = -\sum_{i=1}^{n+1} a_i b_{n+1-i}$. Since a_0 is a unit, we can invert it to get $-b_0 \sum_{i=1}^{n+1} a_i b_{n+1-i}$. This completes the construction of the inverse of f.

2. Let f be such that $f^n = 0$. Then $a_0^n = 0$ and so a_0 is nilpotent. Now suppose for induction that a_i are nilpotent for all $i \leq m$ with exponents n_i. Since the nilradical is an ideal, we can define $g = f - \sum_{i \leq m} a_i x^i$ and g will be nilpotent, say $g^k = 0$. Then $a_{m+1}^k = 0$, and so a_{m+1} is nilpotent.

   The converse does not hold, as we could have a power series where each coefficient is nilpotent, but the exponent for it to go to zero is unbounded. TODO make an actual example.

3. f is in the Jacobson radical of $A[[x]]$ iff $1 - fg$ is a unit for all $g \in A[[X]]$. This is true iff $1 - a_0 b_0$ is a unit for all $b_0 \in A$ by part 1. And finally this is true iff a_0 is in the Jacobson radical of A.

4. Since x is in the Jacobson radical, it is it $\mathfrak{m}$. Then since $A[[x]]/(x) = A$, we see that $\mathfrak{m}/(x) = \mathfrak{m}^c$. Hence $\mathfrak{m}$ is generated by $\mathfrak{m}^c$ and x. Moreover $A[[x]]/\mathfrak{m} = (A[[x]]/(x))/(\mathfrak{m}/(x)) = A/\mathfrak{m}^c$ and so $\mathfrak{m}^c$ is maximal.

5. Let $\mathfrak{p}$ be a prime ideal of A. Then $\mathfrak{p}^e + (x)$ is an ideal of $A[[x]]$ and moreover $A[[x]]/(\mathfrak{p}^e + (x)) = A/\mathfrak{p}$ so $\mathfrak{p}^e + (x)$ is prime. Moreover $\mathfrak{p}$ is the contraction of $\mathfrak{p}^e + (x)$.

Exercise (6). A ring A is such that every ideal not contained in the nilradical contains a nonzero idempotent (that is, an element such that $e^2 = e \neq 0$). Prove that the nilradical and the Jacobson radical of A are equal.

Solution: The nilradical is contained in the Jacobson radical as every maximal ideal is prime, so we just need to show that the opposite inclusion holds. Suppose a is in the Jacobson radical but not the nilradical. Then (a) is an ideal not contained in the nilradical, and so has a nonzero idempotent $e = ab$ (for some $b \in A$). Note also that since a is in the Jacobson radical, $1 - e$ is a unit with inverse c, so $c - ce = 1$. Then $1 - ce = c$, so $1 = (1 - e)(1 - ce) = 1 - e - ce + ce^2 = 1 - e$. This contradicts e being nonzero! So the Jacobson radical and nilradical are equal.

Exercise (7). Let A be a ring in which every element x satisfies $x^n = x$ for some $n > 1$ (depending on x). Show that every prime ideal is maximal.

Solution: Let $\mathfrak{p}$ be prime. Let $x + \mathfrak{p}$ be a nonzero element. Then $x(x^{n-1} - 1) + \mathfrak{p} = 0 + \mathfrak{p}$, so $x^{n-1} \equiv 1 \pmod{\mathfrak{p}}$. Hence $x + \mathfrak{p}$ is a unit, so $\mathfrak{p}$ is maximal.

Exercise (8). Let A be a ring $\neq 0$. Show that the set of prime ideals of A has minimal elements with respect to inclusion.

Solution: We use Zorn's lemma. Firstly, we know that the set of prime ideals is non-empty since it has a maximal ideal. Let C be a chain in the poset of prime ideals. The intersection $\mathfrak{p} = \cap C$ of this chain is an ideal. Suppose $a, b \notin \mathfrak{p}$. Then there is some $\mathfrak{p}_c \in C$ such that $a, b \notin \mathfrak{p}_c$. Hence, $ab \notin \mathfrak{p}_c$, so $ab \notin \mathfrak{p}$. Thus $\mathfrak{p}$ is prime. Every chain has a lower bound, so Zorn's lemma tells us that there is a minimal element.

Exercise (9). Let $\mathfrak{a}$ be an ideal $\neq (1)$ in a ring A. Show that $\mathfrak{a} = \sqrt{\mathfrak{a}} \iff \mathfrak{a}$ is an intersection of prime ideals.

Solution: If $\mathfrak{a} = \sqrt{\mathfrak{a}}$ then $\mathfrak{a}$ is the intersection of all ideals containing it. Conversely, suppose $\mathfrak{a}$ is an intersection of prime ideals $\mathfrak{p}_i$. Let $x \in \sqrt{\mathfrak{a}}$ so that $x^n \in \mathfrak{a}$. Then for each $\mathfrak{p}_i$, $x^n \in \mathfrak{p}_i$ so $x \in \mathfrak{p}_i$. Hence $x \in \mathfrak{a}$.

Exercise (10). Let A be a ring, $\mathfrak{R}$ its nilradical. Show that the following are equivalent:

1. A has exactly one prime ideal;
2. every element of A is either a unit or nilpotent;
3. $A/\mathfrak{R}$ is a field.

Solution: If A has exactly one prime ideal, it is maximal, since A has to have at least one maximal ideal. Moreover, the nilradical is the intersection of all prime ideals, and hence is this ideal. Since every non unit in A is contained in a maximal ideal, which is the nil radical, every non unit is nilpotent.

Now suppose every element is either a unit or nilpotent. Recall that $A/\mathfrak{R}$ has no nilpotent elements. Let $x \in A/\mathfrak{R}$ be nonzero. Then $x \notin A/\mathfrak{R}$, and hence is a unit in A. Thus $A/\mathfrak{R}$ is a field.

Finally suppose $A/\mathfrak{R}$ is a field. Then $\mathfrak{R}$ is maximal. However, it is equal to the intersection of all prime ideals. Thus it is the only prime ideal in A.

Exercise (11). A ring A is *Boolean* if $x^2 = x$ for all $x \in A$. In a Boolean ring A, show that

1. $2x = 0$ for all $x \in A$;
2. every prime ideal $\mathfrak{p}$ is maximal, and $A/\mathfrak{p}$ is a field of two elements;
3. every finitely generated ideal is principal.

Solution:

1. If $x^2 = x$ then $x + 1 = (x + 1)^2 = x^2 + 2x + 1 = x + 2x + 1$. Hence $2x = 0$.
2. Every prime is maximal from Exercise (7). Let $\mathfrak{p}$ be such a prime. Let $x + \mathfrak{p}$ be nonzero. Then $x^2 - x \in \mathfrak{p}$. Hence $x - 1 \in \mathfrak{p}$. Thus $x \equiv 1 \pmod{\mathfrak{p}}$, so $A/\mathfrak{p} \cong \mathbb{F}_2$.
3. Let $\mathfrak{a}$ be generated by a_1, a_2. Let $a = a_1 + a_2 - a_1a_2$. Then $a_1a = a_1 + a_1a_2 - a_1a_2 = a_1$ and similarly $a_2a = a_2$. Every $x \in \mathfrak{a}$ can be written as $x = x_1a_1 = x_2a_2 = x_1a_1a + x_2a_2a$. Thus $\mathfrak{a}$ is generated by a.

   If $\mathfrak{a}$ is generated by $a_1, \cdots, a_{n-1}, a_n$, by induction the $a_1, \cdots, a_{n-1}$ generate a principal ideal, and so we can repeat the above argument.

Exercise (12). A local ring contains no idempotents $\neq 0, 1$.

Solution: Let $\mathfrak{m}$ be the maximal ideal. Let e be an idempotent in A. If $e \in \mathfrak{m}$ then $1 - e$ is a unit, as $\mathfrak{m}$ is the Jacobson radical. If c is the inverse, then $1 = c - ce$, and then $c = 1 - ce$. Then $1 = (1 - e)(1 - ce) = 1 - e - ce + ce = 1 - e$, so $e = 0$. If $e \notin \mathfrak{m}$, then e is a unit. Since $e^2 - e = e(e - 1) = 0$, we get that $e = 1$.

Exercise (13). Let K be a field and let Σ be the set of irreducible monic polynomials f in one indeterminate with coefficients in K. Let A be the polynomial ring over K generated by indeterminates x_f, one for each $f \in \Sigma$. Let $\mathfrak{a}$ be the ideal of A generated by the polynomials $f(x_f)$ for all $f \in \Sigma$. Show that $\mathfrak{a} \neq (1)$.

Then they explain how this leads to a construction of the algebraic closure of a field! You should read it!

Solution: Since K has no nilpotent elements, Exercise (3) tells us that a polynomial in A is a unit if and only if it is a constant unit. However, every $f(x_f)$ is non-constant, and the sum and product of non-constant polynomials over a field will be non-constant. Hence there are no units in $\mathfrak{a}$.

Exercise (14). In a ring A, let Σ be the set of all ideals in which every element is a zero-divisor. Show that the set Σ has maximal elements and that every maximal element of Σ is a prime ideal. Hence, the set of zero-divisor in A is a union of prime ideals.

Solution: We apply Zorn's lemma. Firstly, the zero ideal is in Σ so it is non-empty. Let C be a chain. Then the union is an ideal. Every element in the union a zero-divisor, so the union is an upper bound of C. Hence maximal elements exist. Let $\mathfrak{p}$ be a maximal element. Assume $ab \in \mathfrak{p}$ but $a, b \notin \mathfrak{p}$. Then $(a) + \mathfrak{p}$ and $(b) + \mathfrak{p}$ are strictly larger then $\mathfrak{p}$ and hence contain non-zero divisors $aa' + p, bb' + p$. However $(aa' + p)(bb' + p) = aba'b' + aa'p + bb'p + p^2$ and this is a zero-divisor. This is a contradiction (if $abc = 0$ then a has annihilator bc)! So $\mathfrak{p}$ is prime.

Exercise (15). Let A be a ring and let X be the set of all prime ideals of A. For each subset E of A, let $V(E)$ denote the set of prime ideals of A which contain E. Prove that

1. if $\mathfrak{a}$ is the ideal generated by E, then $V(E) = V(\mathfrak{a}) = V(\sqrt{\mathfrak{a}})$.
2. $V(0) = X$, $V(1) = \varnothing$.
3. if $(E_i)_{i \in I}$ is any family of subsets of A, then
$$V(\bigcup_{i \in I} E_i) = \bigcap_{i \in I} V(E_i)$$
4. $V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b})$ for any ideals $\mathfrak{a}, \mathfrak{b}$ of A.

These results show that the sets $V(E)$ satisfy the axioms for closed sets in a topological space. The resulting topology is called the *Zariski topology*. The topological space X is called the *prime spectrum* of A, and is written $\mathrm{Spec}(A)$.

Solution:

1. If $E \subseteq \mathfrak{p}$ then $\mathfrak{a} \subseteq \mathfrak{p}$ since $\mathfrak{a}$ is the smallest ideal containing E. Conversely if $\mathfrak{a} \subseteq \mathfrak{p}$ then $E \subseteq \mathfrak{a} \subseteq \mathfrak{p}$. Hence $V(E) = V(\mathfrak{a})$. Next, if $\mathfrak{a} \subseteq \mathfrak{p}$ then $\sqrt{\mathfrak{a}} \subseteq \sqrt{\mathfrak{p}} = \mathfrak{p}$. Conversely if $\sqrt{\mathfrak{a}} \subseteq \mathfrak{p}$ then $\mathfrak{a} \subseteq \sqrt{\mathfrak{a}} \subseteq \mathfrak{p}$. Hence $V(\mathfrak{a}) = V(\sqrt{\mathfrak{a}})$.
2. Every prime ideal contains 0 and no prime ideals contain 1 as they are proper. Hence $V(0) = X$ and $V(1) = \varnothing$.
3. $\bigcup_{i \in I} E_i \subseteq \mathfrak{p}$ iff $E_i \subseteq \mathfrak{p}$ for each $i \in I$. Hence $V(\bigcup_{i \in I} E_i) = \bigcap_{i \in I} V(E_i)$.
4. If $\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{p}$ then $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{p}$. Conversely, if $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{p}$ then either $\mathfrak{a} \subseteq \mathfrak{p}$ or $\mathfrak{b} \subseteq \mathfrak{p}$, and hence $\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{p}$. Hence $V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b})$. Next, if $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{p}$ then (as stated) either $\mathfrak{a} \subseteq \mathfrak{p}$ or $\mathfrak{b} \subseteq \mathfrak{p}$. Conversely if $\mathfrak{a} \subseteq \mathfrak{p}$ or $\mathfrak{b} \subseteq \mathfrak{p}$, then $\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{p}$. Hence $V(\mathfrak{a}\mathfrak{b}) \subseteq V(\mathfrak{a}) \cup V(\mathfrak{b})$ and $V(\mathfrak{a}) \cup V(\mathfrak{b}) \subseteq V(\mathfrak{a} \cap \mathfrak{b})$.

Exercise (16). Draw pictures of $\mathrm{Spec}(\mathbb{Z}), \mathrm{Spec}(\mathbb{R}), \mathrm{Spec}(\mathbb{C}[x]), \mathrm{Spec}(\mathbb{R}[x]), \mathrm{Spec}(\mathbb{Z}[x])$.

Solution: TODO i should REALLY come back to this question and describe their pictures...

Exercise (17). For each $f \in A$, let X_f denote the complement of $V(f)$ in $X = \mathrm{Spec}(A)$. The sets X_f are open. Show that they form a basis of open sets for the Zariski topology, and that

1. $X_f \cap X_g = X_{fg}$;
2. $X_f = \varnothing \iff f$ is nilpotent;
3. $X_f = X \iff f$ is a unit;
4. $X_f = X_g \iff \sqrt{(f)} = \sqrt{(g)}$;
5. X is quasi-compact (that is, every open covering of X has a finite subcovering).
6. More generally, each X_f is quasi-compact.
7. An open subset of X is quasi-compact if and only if it is a finite union of sets X_f.

The sets X_f are called *basic open sets* of $X = \mathrm{Spec}(A)$.

Solution: Note that $\mathfrak{p} \in X_f$ iff $f \notin \mathfrak{p}$. Every $\mathfrak{p} \in X$ is proper, and so some $f \in X - \mathfrak{p}$, so $\mathfrak{p} \in X_f$. Hence X_f covers X. Next, if $\mathfrak{p} \in X_f \cap X_g$, then $\mathfrak{p} \in X_{fg}$ as we will show next. Hence the X_f form a basis!

1. Note that since $\mathfrak{p}$ is prime, $f, g \notin \mathfrak{p}$ iff $fg \notin \mathfrak{p}$. Hence $X_f \cap X_g = X_{fg}$ directly!
2. $X_f = \varnothing$ iff $f \in \mathfrak{p}$ for every $\mathfrak{p} \in \operatorname{Spec}(A)$, iff f is in the nilradical.
3. If $X_f = X$ then f is not contained in any prime ideal. Hence it is not contained in any maximal ideal, so it is a unit. Conversely, suppose f is a unit. Then $(f) = A$, so $X_f = X$.
4. If $X_f = X_g$ then $f \in \mathfrak{p}$ iff $g \in \mathfrak{p}$. Recall that $\sqrt{(f)}$ is the intersection of all prime ideals containing (f). But a prime contains (f) iff it contains (g), so $\sqrt{(f)} = \sqrt{(g)}$. Conversely, if $\sqrt{(f)} = \sqrt{(g)}$, then $V(f) = V(\sqrt{(f)}) = V(\sqrt{(g)}) = V(g)$.
5. Let X be covered by $(X_{f_i})_{i \in I}$s. Then $\bigcap_{i \in I} V(f_i) = V(\sum_{i \in I}(f_i)) = \varnothing$ by Exercise (15). Since every maximal ideal is prime, this necessarily implies that $\sum_{i \in I}(f_i) = (1)$. Hence $1 = \sum_{j \in J} f_j g_j$ for some finite J. Then $\sum_{j \in J}(f_j) = 1$, so X is covered by $(X_{f_j})_{j \in J}$.
6. Let X_f be covered by $(X_{fg_i})_{i \in I}$. Then $V(\sum_{i \in I}(fg_i)) = V(f)$. Hence a prime ideal contains f iff it contains $\sum_{i \in I}(fg_i)$. Hence their radicals are equal. Hence $(f) \subseteq \sqrt{\sum_{i \in I}(fg_i)}$. This means that $f^n = \sum_{j \in J} fg_j h_j$ for some finite set j and some n. If $\mathfrak{p}$ is a prime ideal containing $\sum j \in J(fg_j)$, then $f^n \in \mathfrak{p}$, so $f \in \mathfrak{p}$. Hence $V(\sum_{j \in J}(fg_j)) \subseteq V(f)$. Thus $(X_{fg_j})_{j \in J}$ covers X_f.
7. Suppose $U \subseteq X$ is quasi-compact. Take $(X_{f_i})_{i \in I}$ so that $U = \bigcup_{i \in I} X_{f_i}$ from the basis. Then there is a finite subcover $(X_{f_j})_{j \in J}$. Hence U is a finite union of X_f sets.

   The converse is just a fact in topology, that the finite union of quasi-compact sets is quasi-compact. Indeed take a finite set $(X_{f_j})_{j \in J}$. Let $U = \bigcup_{j \in J} X_{f_j}$ and take a cover $(X_i)_{i \in I}$. Then the (X_i) cover each X_{f_j}, and hence each have a finite subcover. Their union is finite and covers U.

Exercise (18). For psychological reasons it is sometimes convenient to denote a prime ideal of A by a letter such as x or y when thinking of it as a point of $X = \operatorname{Spec}(A)$. When thinking of x as a prime ideal of A, we denote it by $\mathfrak{p}_x$ (logically, of course, it is the same thing). Show that

1. the set $\{x\}$ is closed (we say that x is a closed point) in $\operatorname{Spec}(A) \iff \mathfrak{p}_x$ is maximal;
2. $\overline{\{x\}} = V(\mathfrak{p}_x)$;
3. $y \in \overline{\{x\}} \iff \mathfrak{p}_x \subseteq \mathfrak{p}_y$;
4. X is a T_0-space (this means that if x, y are distinct points of X, then either there is a neighbourhood of x which does not contain y, or else there is a neighbourhood of y which does not contain x.

Solution:

1. If $\{x\}$ is closed then $\{x\} = V(\mathfrak{a})$ for some $\mathfrak{a}$. The set of prime ideals containing $\mathfrak{a}$ is just x, so necessarily $\mathfrak{a}$ is maximal and equal to $\mathfrak{p}_x$. Conversely if $\mathfrak{p}_x$ is maximal, then the set of ideals containing it is just $V(\mathfrak{p}_x) = \{\mathfrak{p}_x\} = \{x\}$.

2. A point y is in the closure of $\{x\}$ iff every open neighbourhood of y contains x. This is true iff it is true for every basis element. That is, for all $f \in A$, if $y \in X_f$ then $x \in X_f$. This is equivalent to that, if $f \notin \mathfrak{p}_y$ then $f \notin \mathfrak{p}_x$. Finally this is equivalent to saying $\mathfrak{p}_x \subseteq \mathfrak{p}_y$, i.e. $\mathfrak{p}_y \in V(\mathfrak{p}_x)$.

3. This is the definition of $V(\mathfrak{p}_x)$.

4. Notice that for distinct x, y, from part 3. that if $y \in \overline{\{x\}}$ then $x \notin \overline{\{y\}}$, and so we can take the open set $X - \overline{\{y\}}$ which contains x but not y. If $y \notin \overline{\{x\}}$, then $\mathfrak{p}_x \not\subseteq \mathfrak{p}_y$, hence $\mathfrak{p}_y \notin V(\mathfrak{p}_x)$. Hence we can take the open set $X - V(\mathfrak{p}_x)$ which contains y but not x.

Exercise (19). A topological space X is said to be *irreducible* if $X \neq \varnothing$ and if every pair of non-empty open sets of X intersect, or equivalently if every non-empty open set is dense in X. Show that $\mathrm{Spec}(A)$ is irreducible if and only if the nilradical of A is a prime ideal.

Solution: Note that X is irreducible iff every pair of non-empty open basis elements intersect. Equivalently, for all non-nilpotent $f, g \in A$, the product fg is not nilpotent by Exercise (17). This is exactly the statement of the nilradical being prime!

Exercise (20). Let X be a topological space.

1. If Y is an irreducible (Exercise (19)) subspace of X, then the closure $\overline{Y}$ of Y in X is irreducible.

2. Every irreducible subspace of X is contained in a maximal irreducible subspace.

3. The maximal irreducible subspaces of X are closed and cover X. They are called the *irreducible components* of X. What are the irreducible components of a Hausdorff space?

4. If A is a ring and $X = \mathrm{Spec}(A)$, then the irreducible components of X are the closed sets $V(\mathfrak{p})$, where $\mathfrak{p}$ is a minimal prime ideal of A (Exercise (8)).

Solution:

1. Let U, V be non-empty open subsets of $\overline{Y}$. Then $U \cap Y, V \cap Y$ are non-empty. Hence they interesct, so U and V intersect.

2. If there are no irreducible subspaces of X then the proposition holds trivially. Else, we can apply Zorn's lemma. Let C be a chain of irreducible subspaces. Let U, V be open subsets of the union of C. Then there is some subspace Y containing both U and V. Hence, they intersect, so they intersect in the union. Therefore the union of C is irreducible, and thus is an upper bound of C.

3. If Y is a maximal irreducible subspace, then $\overline{Y}$ is irreducible, and hence equal to Y, so Y is closed. The maximal irreducible subspaces cover X since irreducible spaces cover X, indeed each point is irreducible and contained in some maximal irreducible subspace (this is given by Zorn's lemma).

   Since every pair of points in a Hausdorff space can be separated by disjoint open neighbourhoods, if a subspace has at least two points it is not irreducible. Hence the irreducible subspaces (and irreducible components) are the points, which indeed are closed!

4. Let Y be a closed irreducible subspace. Then it is closed, so it is equal to $V(\sqrt{\mathfrak{a}})$ for some ideal $\mathfrak{a}$. We first aim to show that $\sqrt{\mathfrak{a}}$ is prime. Suppose $fg \in \sqrt{\mathfrak{a}}$. Then $X_{fg} \cap Y = \emptyset$ so one of X_f or X_g is empty in Y. Assume X_f is. Then every prime ideal in Y contains f. Hence the intersection of said prime ideals contains f, which is $\sqrt{\mathfrak{a}}$. If instead X_g were empty, then g would be an element of $\sqrt{\mathfrak{a}}$. Hence $\sqrt{\mathfrak{a}}$ is prime. Conversely, consider $Y = V(\mathfrak{p})$ for some prime $\mathfrak{p}$ and let X_{fg} be empty in Y. Then $fg \in \mathfrak{p}$. Hence either $f \in \mathfrak{p}$ or $g \in \mathfrak{p}$, so either X_f is empty in Y or X_g is, so Y is irreducible.

   Now suppose that $Y = V(\mathfrak{p})$ is an irreducible component. Then if $Y \subseteq Y' = V(\mathfrak{q})$ for some prime $\mathfrak{q}$, then $\mathfrak{q} \subseteq \mathfrak{p}$. However, $\mathfrak{p} = \mathfrak{q}$ since Y is maximal, so $\mathfrak{p}$ is minimal.

Exercise (21). Let $\phi : A \to B$ be a ring homomorphism. Let $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(B)$. If $\mathfrak{q} \in Y$ then $\phi^{-1}(\mathfrak{q})$ is a prime ideal of A, i.e. a point of X. Hence *phi* induces a mapping $\phi^* : Y \to X$. Show that

1. If $f \in A$ then $\phi^{*-1}(X_f) = Y_{\phi(f)}$, and hence ϕ^* is continuous.
2. If $\mathfrak{a}$ is an ideal of A, then $\phi^{*-1}(V(\mathfrak{a})) = V(\mathfrak{a}^e)$.
3. If $\mathfrak{b}$ is an ideal of B, then $\overline{\phi^*(V(\mathfrak{b}))} = V(\mathfrak{b}^c)$.
4. If ϕ is surjective, then ϕ^* is a homeomorphism of Y onto then closed subset $V(\ker(\phi))$ of X. (In particular, $\mathrm{Spec}(A)$ and $\mathrm{Spec}(A/\mathfrak{R})$ (where $\mathfrak{R}$ is the nilradical of A are naturally homeomorphic).
5. If ϕ is injective, then $\phi^*(Y)$ is dense in X. More precisely, $\phi^*(Y)$ is dense in $X \iff \ker(\phi) \subset \mathfrak{R}$.
6. Let $\psi : B \to C$ be another ring homomorphism. Then $(\psi \circ \phi)^* = \phi^* \circ \psi^*$.
7. Let A be an integral domain with just one non-zero prime ideal $\mathfrak{p}$, and K be the field of fractions of A. Let $B = A/\mathfrak{p} \times K$. Define $\phi : A \to B$ by $\phi(x) = (\bar{x}, x)$, where $\bar{x}$ is the image of x in $A/\mathfrak{p}$. Show that ϕ^* is bijective but not a homeomorphism.

Solution:

1. A prime ideal $\mathfrak{p}$ is in $\phi^{*-1}(X_f)$ if $\phi^{-1}(\mathfrak{p})$ does not contain f. This is true iff $\phi(f)$ is not contained in $\mathfrak{p}$, that is, $\mathfrak{p} \in Y_{\phi(f)}$.
2. If $\mathfrak{p} \in \phi^{*-1}(V(\mathfrak{a}))$ then $\mathfrak{a} \subseteq \mathfrak{p}^c$. Hence $\mathfrak{a}^e \subseteq \mathfrak{p}^{ce} \subseteq \mathfrak{p}$, so $\mathfrak{p} \in V(\mathfrak{a}^e)$. Conversely, if $\mathfrak{p} \in V(\mathfrak{a}^e)$, then $\phi^*(\mathfrak{p}) = \mathfrak{p}^c \supseteq \mathfrak{a}^{ec} \supseteq \mathfrak{a}$, so $\mathfrak{p} \in \phi^{*-1}(V(\mathfrak{a}))$.
3. Let $\mathfrak{p} \in \overline{\phi^*(V(\mathfrak{b}))}$. Suppose $f \notin \mathfrak{p}$. Then, $X_f \cap \phi^*(V(\mathfrak{b})))$ is non-empty. Hence there is some $\mathfrak{q} \in V(\mathfrak{b})$ so that $f \notin \mathfrak{q}^c$. Hence $f \notin \mathfrak{b}^c$, so $\mathfrak{b}^c \subseteq \mathfrak{p}$. Conversely, let $\mathfrak{p} \in V(\mathfrak{b}^c)$. Then $\sqrt{\mathfrak{b}^c} = (\sqrt{\mathfrak{b}})^c \subseteq \mathfrak{p}$. Let X_f be a neighbourhood of $\mathfrak{p}$. Then $f \notin (\sqrt{\mathfrak{b}})^c$, and so there is some prime ideal $\mathfrak{q}$ containing $\mathfrak{b}$ so that $f \notin \mathfrak{q}^c$. Hence $X_f \cap \phi^*(V(\mathfrak{b})) \neq \varnothing$, and $\mathfrak{p} \in \overline{\phi^*(V(\mathfrak{b}))}$.
4. We can identify $B = A/\mathfrak{a}$ for some ideal $\mathfrak{a} = \ker(\phi)$. Then ϕ^* is a bijection onto $V(\mathfrak{a})$ as it is an order isomorphism between the prime ideals of A containing $\mathfrak{a}$ and of $A/\mathfrak{a}$. Let Y_f be open in Y. Then there is some f' so that $\phi(f') = f$. We claim $\phi * (Y_f) = X_{f'}$. Indeed if $\mathfrak{q}^c \in \phi^*(Y_f)$, then $f' \notin \mathfrak{q}^c$. Conversely if $\mathfrak{q} \in X'_f$ then $(\mathfrak{q}/\mathfrak{a})^c = \mathfrak{q}$, so $\mathfrak{q}/\mathfrak{a}$ does not contain f.

5. If ϕ is injective, then $\overline{\phi^*(Y)} = \overline{\phi^*(V(0))} = V(0^c) = V(\ker(\phi)) = V(0)$. If $\ker(\phi) \subseteq \mathfrak{R}$ we can identify X with $Spec(A/\ker(\phi))$ in which case ϕ will be injective. Conversely if $\overline{\phi^*(Y)} = X$ then $X = V(0^c) = V(\ker(\phi))$. Hence every prime $\mathfrak{p} \in X$ has $\ker(\phi) \subseteq \mathfrak{p}$, and so $\ker(\phi) \subseteq \mathfrak{R}$.

6. If $\mathfrak{p} \in \mathrm{Spec}(C)$ then $(\psi \circ \phi)^*(\mathfrak{p}) = (\psi \circ \phi)^{-1}(\mathfrak{p}) = \phi^{-1}(\psi^{-1}(\mathfrak{p})) = (\phi^* \circ \psi^*)(\mathfrak{p})$.

7. Firstly $\mathrm{Spec}(A) = \{(0), \mathfrak{p}\}$ has topology $\tau_{\mathrm{Spec}(A)} = \{\varnothing, \mathrm{Spec}(A), \{(0)\}\}$ (think of the closed sets). Notice that this space is connected. Next, $\mathrm{Spec}(B) = \{(0) \times K, A/\mathfrak{p} \times (0)\}$ has topology $\tau_{\mathrm{Spec}(B)} = \{\varnothing, \mathrm{Spec}(B), \{(0) \times K\}, \{A/\mathfrak{p} \times (0)\}\}$. This space is disconnected, so $\mathrm{Spec}(A)$ and $\mathrm{Spec}(B)$ are not homeomorphic. However, ϕ^* is a bijection, since $\phi^*((0) \times K) = \mathfrak{p}$ and $\phi^*(A/\mathfrak{p} \times (0)) = (0)$.

Exercise (22). Let $A = \prod_{i=1}^n A_i$ be the direct product of rings A_i. Show that $\mathrm{Spec}(A)$ is the disjoint union of open (and closed) subspaces X_i, where X_i is canonically homeomorphic to $\mathrm{Spec}(A_i)$. Conversely, let A be any ring. Show that the following are equivalent:

1. $X = \mathrm{Spec}(A)$ is disconnected.

2. $A = A_1 \times A_2$ where neither rings A_1, A_2 is the zero ring.

3. A contains an idempotent $\neq 0, 1$,

In particular, the spectrum of a local ring is always connected (Exercise (12)).

Solution: Let $A = \prod_{i=1}^n A_i$. Since the product of rings is an integral domain iff one of the rings is the integral domain and the rest are zero, and since products commute with quotients, a prime ideal of A is precisely a product of A_is except for one component, which is a prime ideal. We will show that $X = \mathrm{Spec}(A)$ satisfies the universal property of the disjoint union of X_i. Let $f_i : X_i \to Y$ be a choice of continuous maps, we aim to show there is a unique $f : X \to Y$ such that it commutes with the f_i maps through the maps $\pi_i^* : X_i \to X$ induced by the quotient map. Let $x \in X$, then $x = A_1 \times \cdots \times \mathfrak{q} \times \cdots \times A_n$ for a prime ideal $\mathfrak{q}$ of A_i. Hence $x = \pi_i^*(q)$, so $f(x) = f_i(\mathfrak{q})$ is the unique choice of mapping that we can make. It is continuous, as if U is open in Y, then $f^{-1}(U) = \bigcup_i f_i^{-1}(U)$ which is open.

Next, suppose $X = \mathrm{Spec}(A)$ is disconnected. Let $V(\mathfrak{a}), V(\mathfrak{b})$ (for proper ideals $\mathfrak{a}, \mathfrak{b}$ of A) be disjoint clopen subsets of X so that $V(\mathfrak{a}) \cup V(\mathfrak{b}) = X$. Then $V(\mathfrak{a}) \cap V(\mathfrak{b}) = \varnothing = V(A)$, so $\mathfrak{a}$ and $\mathfrak{b}$ are coprime. Also, $X = V(0) = V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{a}\mathfrak{b})$, so $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{R}$. Thus there are $a \in \mathfrak{a}, b \in \mathfrak{b}$ so that $a + b = 1$ and $ab \in \mathfrak{R}$. Then $\sqrt{(a)} + \sqrt{(b)} = (1) = \sqrt{(a^n)} + \sqrt{(b^n)}$ so (a^n) and (b^n) are coprime. Hence there is some $e \in (a^n), e' \in (b^n)$ so that $e + e' = 1$. Then $ee' \in (a^n b^n) = 0$. Finally, $e = e(e + e') = e^2$, so e is an idempotent. It is not 1 since (a^n) is a proper ideal. It is not zero, since then $e' = 1$ and (b^n) is a proper ideal.

Finally, suppose A has an idempontent $e \neq 0, 1$. Then $(1 - e)^2 = 1 - 2e + e^2 = 1 - e$, so $1 - e$ is also idempotent. Then since $e + (1 - e) = 1$, the ideals (e) and $(1 - e)$ are coprime. Also, $e(1 - e) = e - e^2 = 0$, so $(e) \cap (1 - e) = (0)$. Hence by the Chinese remainder theorem, $A \cong A/(e) \times A/(1 - e)$.

Exercise (23). Let A be a Boolean ring (Exercise (11)), and let $X = \mathrm{Spec}(A)$.

1. For each $f \in A$, the set X_f (Exercise (17)) is both open and closed in X.

2. Let $f_1, \ldots, f_n \in A$. Show that $X_{f_1} \cup \cdots \cup X_{f_n} = X_f$ for some $f \in A$.

3. The sets X_f are the only subsets of X which are both open and closed.

4. X is a compact Hausdorff space.

Solution:

1. If $f \notin \mathfrak{p}$ then $1 - f + \mathfrak{p} = 0 + \mathfrak{p}$ since $A/\mathfrak{p}$ is isomorphic to the field of two elements by Exercise (11). Hence $1 - f \in \mathfrak{p}$, so $\mathfrak{p} \in V(1-f)$. Conversely if $\mathfrak{p} \in V(1-f)$ then $1 + \mathfrak{p} = f + \mathfrak{p}$, so $f \notin \mathfrak{p}$ and $\mathfrak{p} \in X_f$. Hence $X_f = V(1-f)$.

2. Observe that
$$\bigcup_{i=1}^{n} X_{f_i} = \bigcup_{i=1}^{n} V(1-f_i) = V\left(\bigcap_{i=1}^{n}(1-f_i)\right).$$
Then $\prod_{i=1}^{n}(1-f_i) \subseteq \bigcap_{i=1}^{n}(1-f_i)$, and this product is finitely generated. Hence, it is principal, equal to (f). Thus $\bigcup_{i=1}^{n} X_{f_i} = V(f)$.

3. Let Y be clopen in X. Then it is quasi-compact by Exercise (17). It is also open, and hence a union $Y = \bigcup_{i \in I} X_{f_i}$. By quasi-compactness, there is a finite subset J of I so that $Y = \bigcup_{j \in J} X_{f_j}$. This it is equal to X_f for some $f \in A$.

4. Let $x, y \in X$ be two disjoint points. Then $x \in V(\mathfrak{p}_x), y \in V(\mathfrak{p}_y)$, and moreover since $\mathfrak{p}_x, \mathfrak{p}_y$ are maximal by Exercise (11), the algebraic sets are singletons. They are also open by above, so they are a separating neighbourhood.

Exercise (24). Let L be a lattice, in which the sup and inf of two elements a, b are denoted by $a \wedge b$ and $a \vee b$ respectively. L is a *Boolean lattice* (or *Boolean algebra*) if

1. L has a least element and a greatest element (denoted $0, 1$ respectively).

2. Each of $\vee, \wedge$ is distributive over the other.

3. Each $a \in L$ has a unique complement $a' \in L$ such that $a \vee a' = 1$ and $a \wedge a' = 0$.

(For example, the set of all of subsets of a set, ordered by inclusion, is a Boolean lattice.)

Let L be a Boolean lattice. Define addition and multiplication in L by the rules
$$a + b = (a \wedge b') \vee (a' \wedge b), \quad ab = a \wedge b.$$
Verify that in this way L becomes a Boolean ring.

Conversely, starting with a Boolean ring A, define an ordering on A as follows: $a \leq b$ means that $a = ab$. Show that, with respect to this ordering, A is a Boolean lattice. In this way, we obtain a one-to-one correspondence between (isomorphism classes of) Boolean rings and (isomorphism classes) of Boolean lattices.

Solution: We have to verify a lot of axioms. Firstly we show de Morgan's law for $\vee$.
$$(a \vee b) \vee (a' \wedge b') = (a \vee b \vee a') \wedge (a \vee b \vee b') = 1 \wedge 1 = 1$$
$$(a \vee b) \wedge (a' \wedge b') = (a \wedge a' \wedge b') \vee (b \wedge a' \wedge b') = 0 \vee 0 = 0$$

So $(a \vee b)' = a' \wedge b'$ by uniqueness of complements. Next we show the de Morgan's law for $\wedge$.

$$(a \wedge b) \vee (a' \vee b') = (a \vee a' \vee b') \wedge (b \vee a' \vee b') = 1 \wedge 1 = 1$$
$$(a \wedge b) \wedge (a' \vee b') = (a \wedge b \wedge a') \vee (b \wedge a' \wedge b') = 0 \vee 0 = 0$$

So $(a \wedge b)' = a' \vee b'$. Next we show a useful fact.

$$a + b = ab' \vee a'b = (a \vee a'b)(b' \vee a'b) = (a \vee a')(a \vee b)(b' \vee a')(b' \vee b) = (a \vee b)(a' \vee b')$$

Now $a + b = (a \vee b)(a' \vee b') = (b \vee a)(b' \vee a') = b + a$ so addition is commutative. Next,

$$\begin{aligned} a(b+c) &= a(b \vee c)(b' \vee c') = \\ &= (ab \vee ac)(b' \vee c') = \\ &= a'ab \vee a'ac \vee (ab \vee ac)(b' \vee c') \\ &= a'(ab \vee ac) \vee (ab \vee ac)(b' \vee c') \\ &= (ab \vee ac)(a' \vee b' \vee c') \\ &= (ab \vee ac)((ab)' \vee (ac)') \\ &= ab + ac \end{aligned}$$

so multiplication distributes over addition. Next,

$$\begin{aligned} a + (b+c) &= (a(b+c)') \vee (a'(b+c)) \\ &= (a(b'c' \vee bc)) \vee a'(b \vee c)(b' \vee c') \\ &= (ab'c' \vee abc) \vee a'(b(b' \vee c') \vee c(b' \vee c')) \\ &= (ab'c' \vee abc) \vee a'(bc' \vee b'c) \\ &= ab'c' \vee abc \vee a'bc' \vee a'b'c \\ &= ab'c' \vee a'bc' \vee a'b'c \vee abc \\ &= (ab' \vee a'b)c' \vee a'b'c \vee abc \\ &= (a(a' \vee b') \vee b(a' \vee b'))c' \vee a'b'c \vee abc \\ &= (a \vee b)(a' \vee b')c' \vee a'b'c \vee abc \\ &= (a \vee b)(a' \vee b')c' \vee (a'b' \vee ab)c \\ &= (a+b)c' \vee (a+b)'c = (a+b) + c \end{aligned}$$

so addition is associative (oh my). Hence this is a ring. It is boolean, as $a^2 = a$.

Conversely, we show A is a Boolean algebra. Firstly, $ab = a \wedge b$ is known from the definition of $\leq$. Next, we suppose $a \vee b = a + b + ab$. If $a \leq b$, then $a = ab$, so $a + ab = 0$. Hence $a + b + ab = a \vee b = b$, so $a \vee b$ is the least upper bound. Next, $0 \leq a$ since $0a = 0$, and $a \leq 1$ since $1a = a$. Next, $a(b \vee c) = ab + ac + abc = ab + ac + abac = ab \vee ac$ and

$$a \vee bc = a + bc + abc = a + ac + ac + ab + bc + abc + aab + abc + abac = (a + b + ab)(a + c + ac) = (a \vee b)(a \vee c)$$

noting $2x = 0$ by Exercise (11), so $\vee$ and $\wedge$ distribute. Finally, for $a \in A$, we define $a' = 1 - a$. Then $aa' = a^2 - a = 0$ and $a \vee a' = a + 1 - a + a^2 - a = 1$, so a' is the complement of a. If $ab = 0$ and $a \vee b = a + b + ab = 1$ then $b = 1 - a = a'$ so a' is unique. Hence A is a Boolean lattice.

We obtain a bijectino between Boolean lattices and Boolean rings as these constructions are inverse. Indeed for a fixed Boolean algebra L, $a \leq b$ iff $a = ab$, and for a fixed Boolean ring A,

$$ab' \vee a'b = ab' + a'b + aa'bb' = ab' + a'b = ab' + 2a'b' + a'b = (a + a')b' + a'(b + b') = a' + b' = a + b$$

and so we recover the same structure going through the composite of each construction in both directions.

Exercise (25). From the last two exercises deduce Stone's theorem, that every Boolean lattice is isomorphic to the lattice of open-and-closed subsets of some compact Hausdorff topological space.

Solution: Let L be a Boolean lattice. Then L has Boolean ring structure by Exercise (24). Hence $X = \mathrm{Spec}(L)$ is compact Hausdorff by Exercise (23). We also have that the clopen subsets of X are precisely the X_a for $a \in L$. From Exercise (17) we have that $X_a \cap X_b = X_{ab}$, so $X_a \subseteq X_b$ iff $a = ab$, i.e. $a \leq b$. Hence the map from L to the clopen sets of X is monotone. Next, we can attempt to define a map from the clopen sets of X to L mapping X_a to a. We have to verify that this map is well defined. If $X_a = X_b$, then $\sqrt{(a)} = \sqrt{(b)}$ by Exercise (17). But in a boolean ring, $x^n = x$, so $(a) = (b)$. Hence $a = b$ as L has no non-identity units ($u = uuu^{-1} = u^{-1}$ and so $u^2 = u = 1$). This map we have now defined is monotone from earlier, so we have an order isomorphism between L and the clopen sets of $\mathrm{Spec}\, L$.

Exercise (26). Let A be a ring. The subspace of $\mathrm{Spec}(A)$ consisting of the *maximal* ideals of A, with the induced topology, is called the *maximal spectrum* of A and is denoted $\mathrm{Max}(A)$. For arbitrary commutative rings it does not have the nice functorial properties of $\mathrm{Spec}(A)$ (see Exercise (21)), because the inverse image of a maximal ideal under a ring homomorphism need not be maximal. Let X be a compact Hausdorff space and let $C(X)$ denote the ring of all real-valued continuous functions on X (add and multiply functions by adding and multiplying their values). For each $x \in X$, let $\mathfrak{m}_x$ be the set of all $f \in C(X)$ such that $f(x) = 0$. The ideal $\mathfrak{m}_x$ is maximal, because it is the kernel of the (surjective) homomorphism $C(X) \to \mathbb{R}$ which takes f to $f(x)$. If $\tilde{X}$ denotes $\mathrm{Max}(C(X))$, we have therefore defined a mapping $\mu : X \to \tilde{X}$, namely $x \mapsto \mathfrak{m}_x$.

We shall show that μ is a homeomorphism of X onto $\tilde{X}$.

1. Let $\mathfrak{m}$ be any maximal ideal of $C(X)$, and let $V = V(\mathfrak{m})$ be the set of common zeros of the functions in $\mathfrak{m}$, that is,

   $$V = \{x \in X : f(x) = 0 \text{ for all } f \in \mathfrak{m}\}.$$

   Suppose that V is empty. Then for each $x \in X$, there exists $f_x \in \mathfrak{m}$ such that $f_x(x) \neq 0$. Since f_x is continuous, there is an open neighbourhood U_x of x in X on which f_x does not vanish. By compactness a finite number of the neighbourhoods, say $U_{x_1}, \ldots, U_{x_n}$ cover X. Let

   $$f = f_{x_1}^2 + \cdots + f_{x_n}^2.$$

   Then f does not vanish at any point of X, hence it is a unit in $C(X)$. But this contradicts $f \in \mathfrak{m}$, hence V is not empty.

2. By Urysohn's lemma (this is the only non-trivial fact required in the argument) the continuous functions separate the points of X. Hence $x \neq y \implies \mathfrak{m}_x \neq \mathfrak{m}_y$, and therefore μ is injective.

3. Let $f \in C(X)$; let
$$U_f = \{x \in X : f(x) \neq 0\}$$
and let
$$\tilde{U}_f = \{\mathfrak{m} \in \tilde{X} : f \notin \mathfrak{m}\}.$$
Show that $\mu(U_f) = \tilde{U}_f$. The open sets U_f (resp. $\tilde{U}_f$) form a basas of the topology of X (resp. of $\tilde{X}$) and therefore μ is a homeomorphism. Thus X can be reconstructed from the ring of functions $C(X)$.

Solution: Firstly if $x \in U_f$ then $\mathfrak{m}_x \in \tilde{U}_f$ as $f + \mathfrak{m} = 0$, so $\mu(U_f) \subseteq \tilde{U}_f$. Next, if $\mathfrak{m} \in \tilde{U}_f$, then $\mathfrak{m} = \mathfrak{m}_x$ for some $x \in X$ by surjectivity of μ. Hence $f(x) \neq 0$, so $x \in U_f$. Thus $\mathfrak{m} = \mu(x)$, so $\mu(U_f) = \tilde{U}_f$.

2 Modules

Exercise 2.2. 1. $\mathrm{Ann}(M + N) = \mathrm{Ann}(M) \cap \mathrm{Ann}(N)$

2. $(N : P) = \mathrm{Ann}((N + P)/N)$

Solution:

1. Note that $a(M + N) = aM + aN$. Hence $a \in \mathrm{Ann}(M + N)$ iff $aM + aN \subseteq 0$ iff $aM \subseteq 0$ and $aN \subseteq 0$ iff $a \in \mathrm{Ann}(M) \cap \mathrm{Ann}(N)$.

2. If $aP \subseteq N$ then every $p \in P$ has $ap \in N$. As $P \subseteq N + P$ this means $ap + N = 0 + N$, so $a \in \mathrm{Ann}((N+P)/N)$. Conversely, suppose $a(n+p)+N = 0+N$ for all $n+p \in N+P$. Then $ap + N = 0 + N$, so $ap \in N$ and so $a \in (N : P)$.

Exercise 2.16. Let A, B be rings, let M be an A-module, P be a B-module and N an (A, B)-bimodule. Then $M \otimes_A N$ is naturally a B-module, $N \otimes_B P$ is naturally an A-module, and we have
$$(M \otimes_A N) \otimes_B P \cong M \otimes_A (N \otimes_B P)$$

Solution: I shall interpret the word naturally to mean that $- \otimes_A N$ is a functor from $\mathbf{Mod}_A$ to $\mathbf{Mod}_B$. It is sufficient to only show this direction of tensoring, the other direction should be identical. For each $b \in B$, define a bilinear map $(m, n) \mapsto m \otimes (bn)$. This induces a map from $M \otimes_A N \to M \otimes_A N$. This multiplication preserves addition and multiplication of the ring structure (indeed $m \otimes (b'(bn)) = m \otimes ((b'b)n)$ and $m \otimes ((b + b')n) = m \otimes (bn + b'n)$). Hence this makes $M \otimes_A N$ a B-module. Now suppose $f : M \to M'$ is an A-module homomorphism. We obtain a map $f \otimes N : M \otimes_A N \to M' \otimes N$ bilinearly mapping $(m, n) \mapsto f(m) \otimes n$. Indeed $\mathrm{id} \otimes N = \mathrm{id}$ and $(g \circ f) \otimes N = (g \otimes N) \circ (f \otimes N)$ (expand the definition).

I shall interpret this big isomorphism we have to prove as an (A, B)-bimodule isomorphism. These tensor products are special cases of a general bimodule tensor product described as follows. If M is an A-module and N is an (A, B)-bimodule, then $M \otimes_A N$ is an (A, B)-bimodule. It has the universal property that maps out of it are naturally isomorphic to A-bilinear maps that are B-linear in the right component. Indeed given such a bilinear map $f : M \times N \to P$, define $\hat{f} : M \otimes_A N \to P$ with A-bilinearity. A bimodule homomorphism is a map that is both an A and

B-module homomorphism, and here $\hat{f}$ is already an A-module homomorphism. To see that it is a B-module homomorphism, observe

$$\hat{f}(b(m \otimes n)) = \hat{f}(m \otimes bn) = f(m, bn) = bf(m, n) = b\hat{f}(m \otimes n)$$

and extend by linearity. This construction is natural in P since post composition does not affect anything we have done. An identical construction can be made for an (A, B)-bimodule N and B-module P where we require A-linearity in the left component.

Now that we have a universal property, we can prove the big isomorphism using the yoneda lemma. To define a map out of $(M \otimes_A N) \otimes_B P$, we have to define a bilinear map out of $(M \otimes_A N) \times P$ that is A-linear in the left component. That is, we have to define a map out of $M \otimes_A N \times P$ that is a bimodule homomorphism with P fixed, and is B-linear with $M \otimes_A N$ fixed. This is the same thing as a map out of $M \times N \times P$ that is A-linear in M, B-linear in P and linear in both A and B in N. Reversing our steps, this shows that the universal property for $(M \otimes_A N) \otimes_B P$ and $M \otimes_A (N \otimes_B P)$ are the same. Hence, they are isomorphic as (A, B)-bimodules.

Exercise 2.21. If $f : A \to B$ is a ring homomorphism and M is a flat A-module, then $M_B = B \otimes_A M$ is a flat B-module.

Solution: Let

$$0 \to N' \to N \to N'' \to 0$$

be an exact sequence of B-modules. Tensoring with M_B gives

$$0 \to (M \otimes_A B) \otimes_B N' \to (M \otimes_A B) \otimes_B N \to (M \otimes_A B) \otimes_B N'' \to 0$$

but this is isomorphic to the following sequence by exercise 2.16.

$$0 \to M \otimes_A (B \otimes_B N') \to M \otimes_A (B \otimes_B N) \to M \otimes_A (B \otimes_B N'') \to 0$$

This sequence is exact as a sequence of A-modules, but since kernels and images of maps are the same when regarded as a sequence of B-modules, this sequence is also exact as a B-module sequence. Hence M_B is flat.

Exercise (1). Show that $(\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) = 0$ if m, n are coprime.

Solution: Let A be a Z-module (or abelian group) and let $f : (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z}) \to A$ be a bilinear map. Then $f(0, 0) = f(m, 0) = mf(1, 0) = f(1, 0)$ and $f(0, 0) = f(0, n) = nf(0, 1) = f(0, 1)$. Hence, the induced map $\hat{f} : (\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) \to A$ is the zero map. Thus every map out of $(\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z})$ is the zero map, and $(\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) \cong 0$.

Exercise (2). Let A be a ring, $\mathfrak{a}$ an ideal, M an A-module. Show that $A/\mathfrak{a} \otimes M$ is isomorphic to $M/\mathfrak{a}M$.

Solution: We will first show that $\mathfrak{a} \otimes M \cong \mathfrak{a}M$. Define $f : \mathfrak{a} \otimes M \to \mathfrak{a}M$ by $a \otimes m \mapsto am$. Similarly define $g : \mathfrak{a}M \to \mathfrak{a} \otimes M$ by $am \mapsto a \otimes m$. Both maps are homomorphisms, and are clearly self inverse.

Next consider the exact sequence

$$0 \to \mathfrak{a} \to A \to A/\mathfrak{a} \to 0.$$

Since tensor is right exact, we obtain the following exact sequence

$$\mathfrak{a}M \to M \to A/\mathfrak{a} \otimes M \to 0.$$

The left map is given from sending $am \mapsto am$ and so is injective. Hence we can extend the sequence to the left with a zero.

$$0 \to \mathfrak{a}M \to M \to A/\mathfrak{a} \otimes M \to 0$$

Thus $A/\mathfrak{a} \otimes M \cong M/\mathfrak{a}M$.